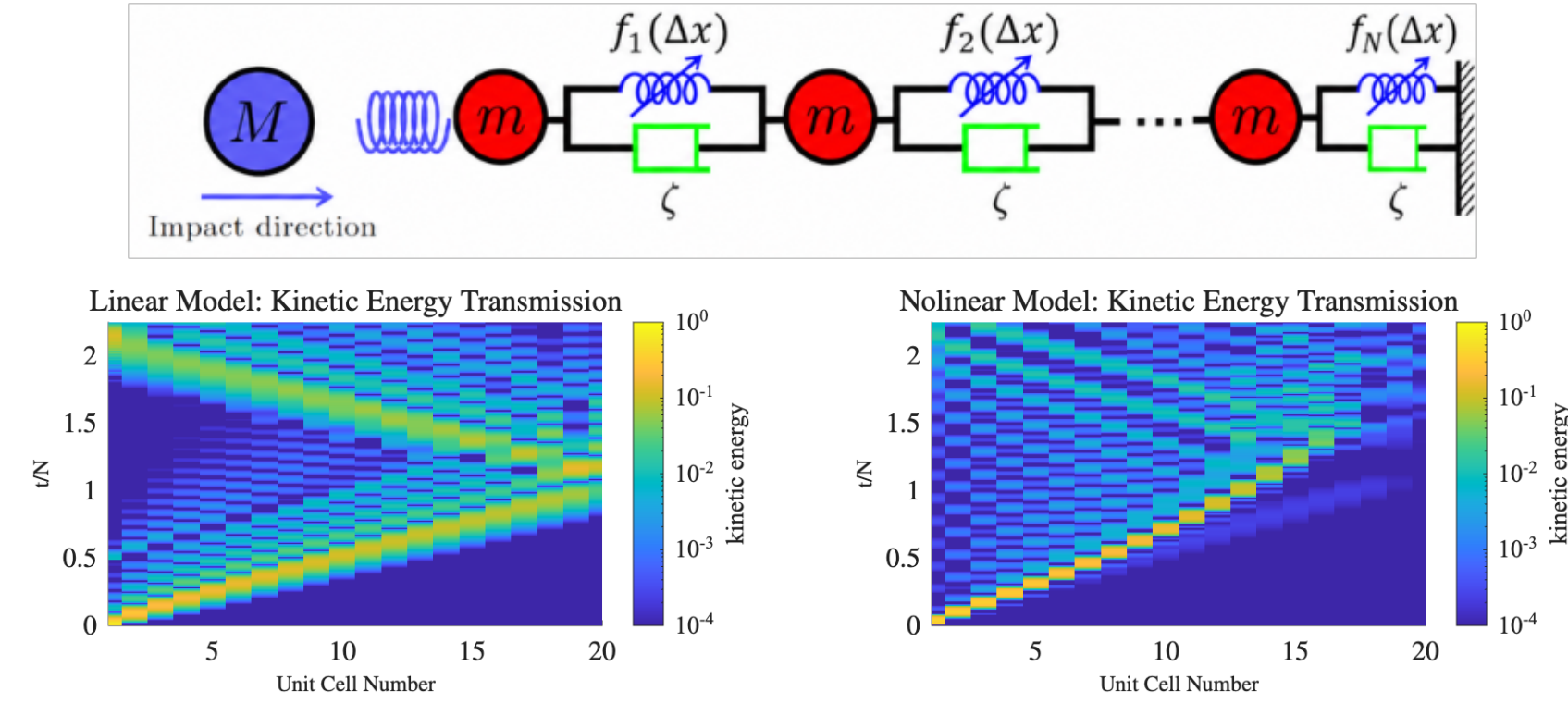


Background and Motivation

- How can nonlinear wave dynamics be exploited in a metamaterial design for impact mitigation?
- Polynomial springs in 1D mass spring damper chains can dissipate kinetic energy for a specific impactor better than linear springs [1].
- However, impact conditions are uncertain. Is there a robust nonlinear spring that can work for a range of impact conditions?



Main Contributions

- Formulated the robust design optimization framework for the Discrete Element Model (DEM).
- Accelerated DEM compute time by porting to Python.
- Exploited MC high throughput structure for parallelization.
- Enabled multinode computing using job arrays on **Dane**.

Robust Design Optimization Problem

The design variables are the 2nd and 3rd coefficients of a polynomial spring force equation,

$$\begin{aligned} f_{\text{non}}(x) &= c_1 x + c_2 x^2 + c_3 x^3, \\ f_{\text{lin}}(x) &= c_1 x, \end{aligned} \quad (1)$$

which are found by solving the optimization problem

$$(c_2^*, c_3^*) = \arg \min_{(c_2, c_3) \in \mathcal{D}} J(c_2, c_3). \quad (2)$$

For the impactor conditions $\mathbf{Z} = \left[\zeta = 0.01, \frac{M}{M_0} = 0.05, \frac{V}{V_0} = 1 \right]$, we formulate the **kinetic energy ratio as the cost function**,

$$J_{\text{det}}(c_2, c_3) = KE_{\text{ratio}}(c_2, c_3; \mathbf{Z}) = \frac{KE_{\text{non}}(c_2, c_3; \mathbf{Z})}{KE_{\text{lin}}(\mathbf{Z})}, \quad (3)$$

$$\text{where } KE = \max_{t \geq 0} \left(\frac{1}{2} m \dot{x}_N^2 \right). \quad (4)$$

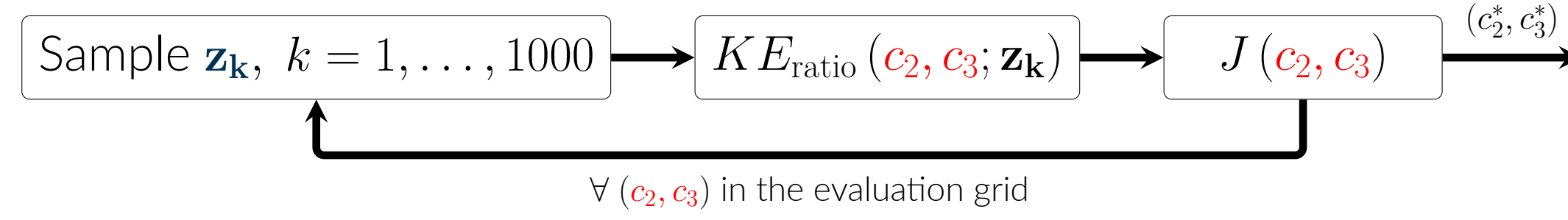
However, the viscosity, impactor velocity and mass are uncertain, so we model them as random variables:

$$\mathbf{Z} = \begin{bmatrix} \zeta \\ \frac{M}{M_0} \\ \frac{V}{V_0} \end{bmatrix} \sim \mathcal{U} \left(\begin{bmatrix} 0.005, 0.025 \\ 0.030, 0.080 \\ 0.500, 1.100 \end{bmatrix} \right). \quad (5)$$

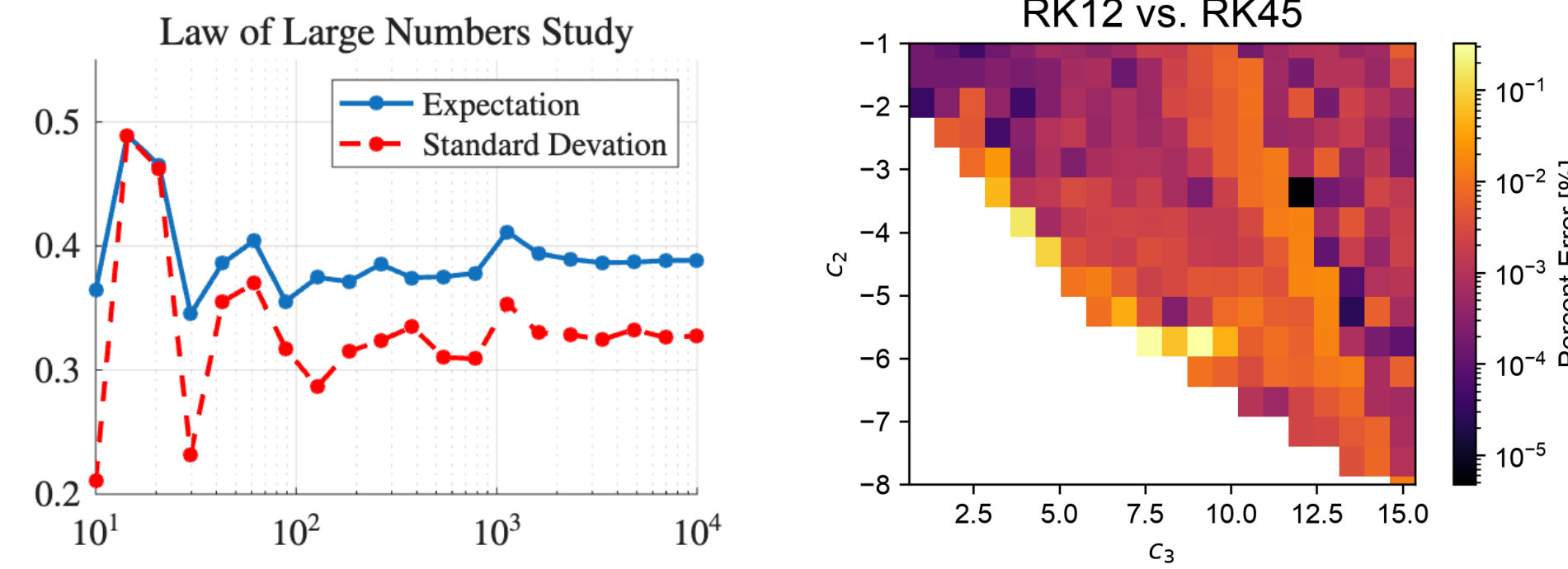
The **cost function for robust optimization** becomes

$$J_{\text{rob}}(c_2, c_3) = \omega_1 \cdot \mathbb{E}[KE_{\text{ratio}}(c_2, c_3; \mathbf{Z})] + \omega_2 \cdot \sqrt{\mathbb{V}[KE_{\text{ratio}}(c_2, c_3; \mathbf{Z})]}. \quad (6)$$

Robust Optimization: Function Evaluations



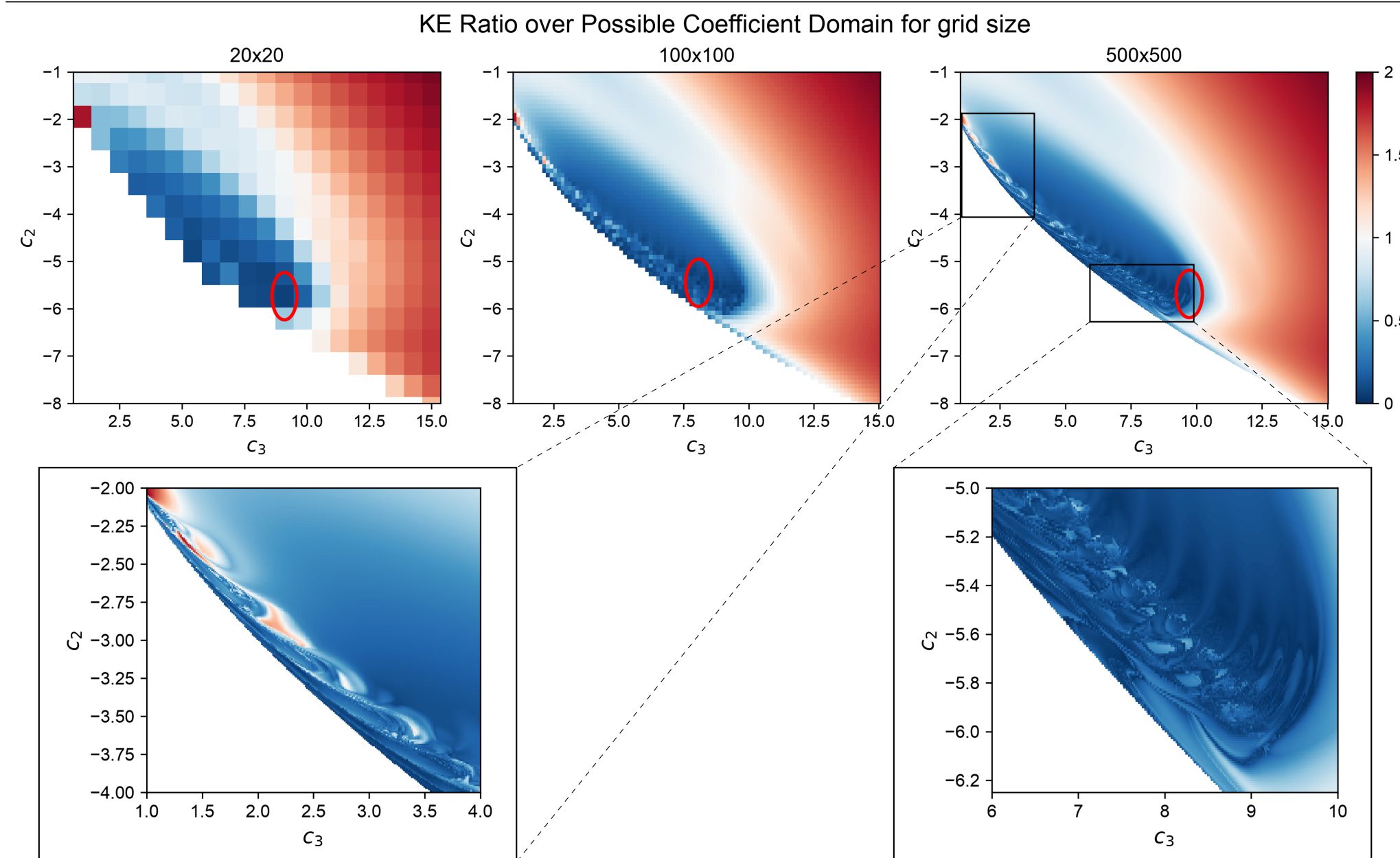
Code Verification



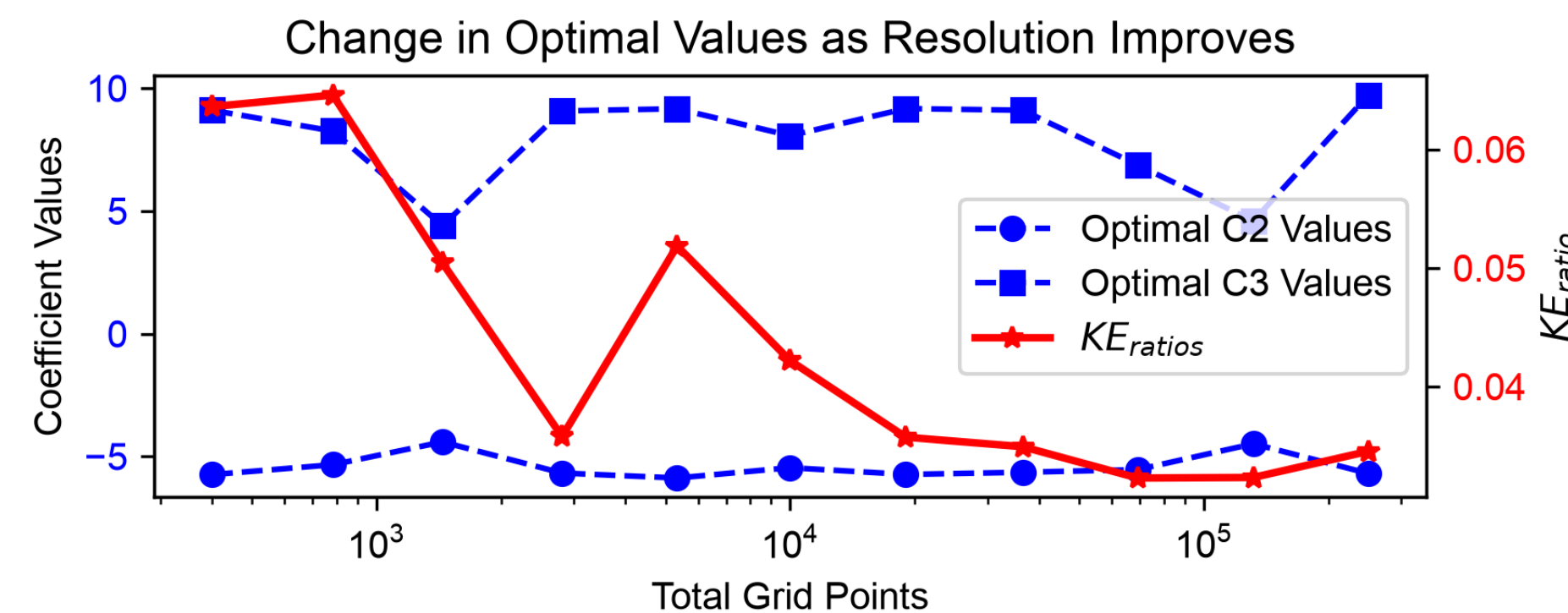
20x20 Grid	Matlab*	Python*
KE_{ratio}	0.0637	0.0636
Optimal Coeffs	$c_2 = -5.74, c_3 = 9.11$	$c_2 = -5.74, c_3 = 9.11$
CPU hrs	0.21	0.05
Wall Clock hrs	0.015	0.004

*Executed on a local machine with 48 GB RAM using 13 parallel CPUs (4.51 GHz)

Deterministic Optimization Results

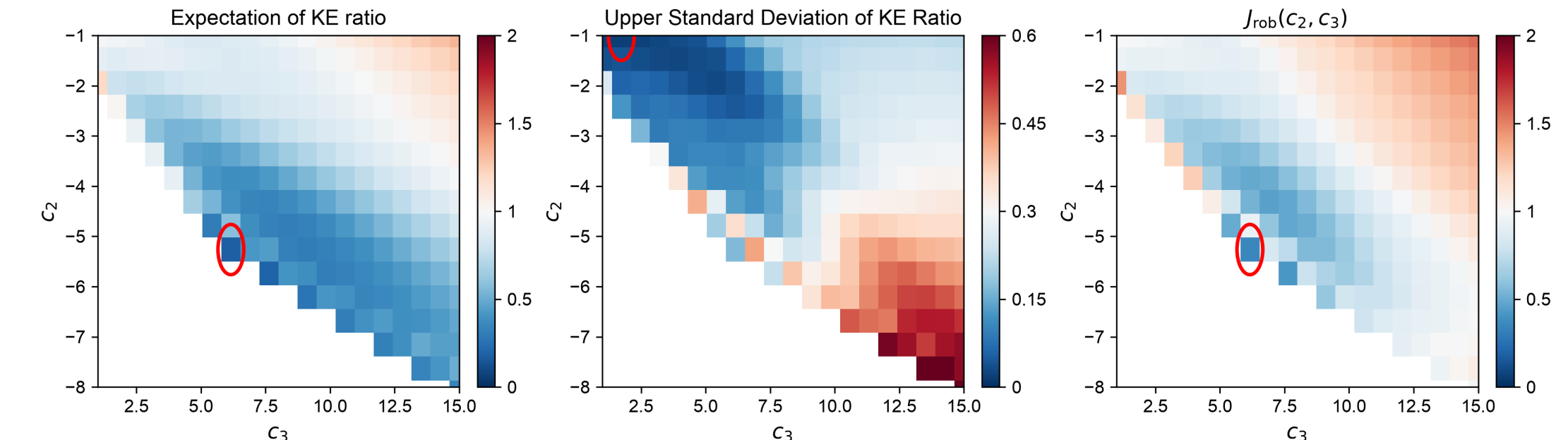


The high-resolution image reveals the highly nonlinear domain, highlighting why gradient solvers struggle to find the global minimum.



Robust Design Optimization Results

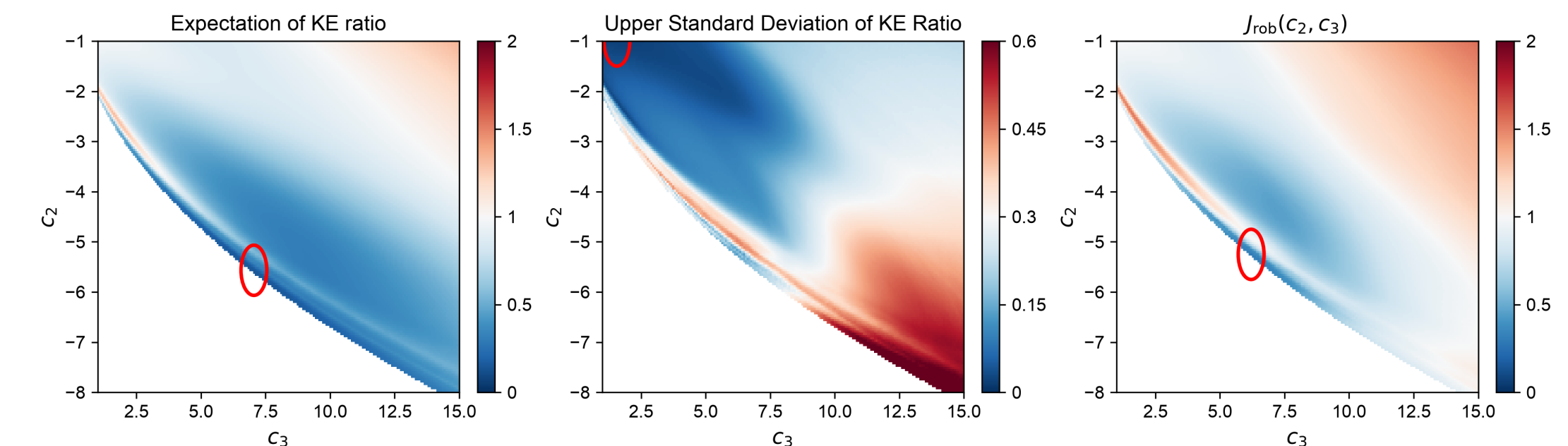
From the Law of Large Numbers, we concluded that 1000 realizations of $\mathbf{Z}$ accurately estimates the mean and standard deviation of the KE_{ratio} .



20x20 Grid	Matlab	Python Local	Python Dane
CPU hrs	45.5	52.5	59
Wall Clock hrs	3.5	5.4	0.75

Computing the 200×200 grid below takes 3h20min on **Dane** with **30 nodes (3360 cores)** and **7443 CPUh** with a CPU efficiency over 80% on all nodes.

An equivalent study in series would take approximately 310 days.



Unlike the deterministic model, the optimal spring coefficients do not change with the grid refinement, indicating that the robustness smooths out the strong nonlinearities in the domain.

Future Work

- Add uncertainty to the coefficients for a true stochastic system.
- Expand to a 2D DEM with more random variables, including impactor size and angle of attack.
- Experimental validation of impact distributions.

Acknowledgements

This work was supported by the Department of Energy, National Nuclear Security Administration (DOE/NNSA) PSAAP IV under Award Number DE-NA0004262.

References

- [1] B MacNider, H Xiu, C Tamur, K Qian, I Frankel, M Brandy, A Kim H, and N Boechler. "Customizable wave tailoring nonlinear materials enabled by bilevel inverse design". In: *Nature Communications* 16.1 (Apr. 10, 2025), p. 3425. ISSN: 2041-1723. DOI: 10.1038/s41467-025-58630-8. (Visited on 09/29/2025).